JavaScript is disabled on your browser.

compact1, compact2, compact3

java.nio

Class FloatBuffer

java.lang.Object
    java.nio.Buffer
        java.nio.FloatBuffer

All Implemented Interfaces:

`Comparable<FloatBuffer>`

```
public abstract class FloatBuffer
extends Buffer
implements Comparable<FloatBuffer>
```

A float buffer.

This class defines four categories of operations upon float buffers:

- Absolute and relative *get* and *put* methods that read and write single floats;
- Relative *bulk get* methods that transfer contiguous sequences of floats from this buffer into an array; and
- Relative *bulk put* methods that transfer contiguous sequences of floats from a float array or some other float buffer into this buffer; and
- Methods for `compacting`, `duplicating`, and `slicing` a float buffer.

Float buffers can be created either by *allocation*, which allocates space for the buffer's content, by *wrapping* an existing float array into a buffer, or by creating a *view* of an existing byte buffer.

Like a byte buffer, a float buffer is either *direct* or *non-direct*. A float buffer created via the `wrap` methods of this class will be non-direct. A float buffer created as a view of a byte buffer will be direct if, and only if, the byte buffer itself is direct. Whether or not a float buffer is direct may be determined by invoking the `isDirect` method.

Methods in this class that do not otherwise have a value to return are specified to return the buffer upon which they are invoked. This allows method invocations to be chained.

Since:

1.4

Method Summary

Instance Methods | Abstract Methods | Concrete Methods

Modifier and Type	Method	Description
static FloatBuffer	allocate(int capacity)	Allocates a new float buffer.
float[]	array()	Returns the float array that backs this buffer *(optional operation)*.
int	arrayOffset()	Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.
abstract FloatBuffer	asReadOnlyBuffer()	Creates a new, read-only float buffer that shares this buffer's content.
abstract FloatBuffer	compact()	Compacts this buffer *(optional operation)*.
int	compareTo(FloatBuffer that)	Compares this buffer to another.
abstract FloatBuffer	duplicate()	Creates a new float buffer that shares this buffer's content.

boolean	**equals**(**Object** ob)	Tells whether or not this buffer is equal to another object.
abstract float	**get**()	Relative *get* method.
FloatBuffer	**get**(float[] dst)	Relative bulk *get* method.
FloatBuffer	**get**(float[] dst, int offset, int length)	Relative bulk *get* method.
abstract float	**get**(int index)	Absolute *get* method.
boolean	**hasArray**()	Tells whether or not this buffer is backed by an accessible float array.
int	**hashCode**()	Returns the current hash code of this buffer.
abstract boolean	**isDirect**()	Tells whether or not this float buffer is direct.
abstract **ByteOrder**	**order**()	Retrieves this buffer's byte order.
abstract **FloatBuffer**	**put**(float f)	Relative *put* method *(optional operation)*.
FloatBuffer	**put**(float[] src)	Relative bulk *put* method *(optional operation)*.
FloatBuffer	**put**(float[] src, int offset, int length)	Relative bulk *put* method *(optional operation)*.
FloatBuffer	**put**(**FloatBuffer** src)	

		Relative bulk *put* method *(optional operation)*.
abstract **FloatBuffer**	**put**(int index, float f)	Absolute *put* method *(optional operation)*.
abstract **FloatBuffer**	**slice**()	Creates a new float buffer whose content is a shared subsequence of this buffer's content.
String	**toString**()	Returns a string summarizing the state of this buffer.
static **FloatBuffer**	**wrap**(float[] array)	Wraps a float array into a buffer.
static **FloatBuffer**	**wrap**(float[] array, int offset, int length)	Wraps a float array into a buffer.

Methods inherited from class java.nio.Buffer

capacity, clear, flip, hasRemaining, isReadOnly, limit, limit, mark, position, position, remaining, reset, rewind

Methods inherited from class java.lang.Object

clone, finalize, getClass, notify, notifyAll, wait, wait, wait

Method Detail

allocate

```
public static FloatBuffer allocate(int capacity)
```

Allocates a new float buffer.

The new buffer's position will be zero, its limit will be its capacity, its mark will be undefined, and each of its elements will be initialized to zero. It will have a `backing array`, and its `array offset` will be zero.

Parameters:

`capacity - The new buffer's capacity, in floats`

Returns:

`The new float buffer`

Throws:

`IllegalArgumentException - If the capacity is a negative integer`

wrap

```
public static FloatBuffer wrap(float[] array,
                               int offset,
                               int length)
```

Wraps a float array into a buffer.

The new buffer will be backed by the given float array; that is, modifications to the buffer will cause the array to be modified and vice versa. The new buffer's capacity will be `array.length`, its position will be `offset`, its limit will be `offset + length`, and its mark will be undefined. Its `backing array` will be the given array, and its `array offset` will be zero.

Parameters:

`array - The array that will back the new buffer`

`offset - The offset of the subarray to be used; must be non-negative and no larger than array.length. The new buffer's position will be set to this value.`

`length - The length of the subarray to be used; must be non-negative and no larger than array.length - offset. The new buffer's limit will be set to offset + length.`

Returns:

`The new float buffer`

Throws:

`IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold`

wrap

```
public static FloatBuffer wrap(float[] array)
```

Wraps a float array into a buffer.

The new buffer will be backed by the given float array; that is, modifications to the buffer will cause the array to be modified and vice versa. The new buffer's capacity and limit will be `array.length`, its position will be zero, and its mark will be undefined. Its `backing array` will be the given array, and its `array offset>` will be zero.

Parameters:

`array - The array that will back this buffer`

Returns:

`The new float buffer`

slice

```
public abstract FloatBuffer slice()
```

Creates a new float buffer whose content is a shared subsequence of this buffer's content.

The content of the new buffer will start at this buffer's current position. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's position will be zero, its capacity and its limit will be the number of floats remaining in this buffer, and its mark will be undefined. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

`The new float buffer`

duplicate

```
public abstract FloatBuffer duplicate()
```

Creates a new float buffer that shares this buffer's content.

The content of the new buffer will be that of this buffer. Changes to this buffer's content will be visible in the new buffer, and vice versa; the two buffers' position, limit, and mark values will be independent.

The new buffer's capacity, limit, position, and mark values will be identical to those of this buffer. The new buffer will be direct if, and only if, this buffer is direct, and it will be read-only if, and only if, this buffer is read-only.

Returns:

`The new float buffer`

asReadOnlyBuffer

```
public abstract FloatBuffer asReadOnlyBuffer()
```

Creates a new, read-only float buffer that shares this buffer's content.

The content of the new buffer will be that of this buffer. Changes to this buffer's content will be visible in the new buffer; the new buffer itself, however, will be read-only and will not allow the shared content to be modified. The two buffers' position, limit, and mark values will be independent.

The new buffer's capacity, limit, position, and mark values will be identical to those of this buffer.

If this buffer is itself read-only then this method behaves in exactly the same way as the `duplicate` method.

Returns:

The new, read-only float buffer

get

```
public abstract float get()
```

Relative *get* method. Reads the float at this buffer's current position, and then increments the position.

Returns:

The float at the buffer's current position

Throws:

`BufferUnderflowException` - If the buffer's current position is not smaller than its limit

put

```
public abstract FloatBuffer put(float f)
```

Relative *put* method *(optional operation)*.

Writes the given float into this buffer at the current position, and then increments the position.

Parameters:

f - The float to be written

Returns:

This buffer

Throws:

`BufferOverflowException` - If this buffer's current position is not smaller than its limit

`ReadOnlyBufferException` - If this buffer is read-only

get

```
public abstract float get(int index)
```

Absolute *get* method. Reads the float at the given index.

Parameters:

`index` - The index from which the float will be read

Returns:

The float at the given index

Throws:

`IndexOutOfBoundsException` - If index is negative or not smaller than the buffer's limit

put

```
public abstract FloatBuffer put(int index,
                                float f)
```

Absolute *put* method *(optional operation)*.

Writes the given float into this buffer at the given index.

Parameters:

`index` - The index at which the float will be written

`f` - The float value to be written

Returns:

This buffer

Throws:

`IndexOutOfBoundsException` - If index is negative or not smaller than the buffer's limit

`ReadOnlyBufferException` - If this buffer is read-only

get

```
public FloatBuffer get(float[] dst,
                       int offset,
                       int length)
```

Relative bulk *get* method.

This method transfers floats from this buffer into the given destination array. If there are fewer floats remaining in the buffer than are required to satisfy the request, that is, if `length > remaining()`, then no floats are transferred and a `BufferUnderflowException` is thrown.

Otherwise, this method copies `length` floats from this buffer into the given array, starting at the current position of this buffer and at the given offset in the array. The position of this buffer is then incremented by `length`.

In other words, an invocation of this method of the form `src.get(dst, off, len)` has exactly the same effect as the loop

```
     for (int i = off; i < off + len; i++)
         dst[i] = src.get():
```

except that it first checks that there are sufficient floats in this buffer and it is potentially much more efficient.

Parameters:

`dst - The array into which floats are to be written`

`offset - The offset within the array of the first float to be written; must be non-negative and no larger than dst.length`

`length - The maximum number of floats to be written to the given array; must be non-negative and no larger than dst.length - offset`

Returns:

`This buffer`

Throws:

`BufferUnderflowException - If there are fewer than length floats remaining in this buffer`

`IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold`

get

```
public FloatBuffer get(float[] dst)
```

Relative bulk *get* method.

This method transfers floats from this buffer into the given destination array. An invocation of this method of the form `src.get(a)` behaves in exactly the same way as the invocation

```
src.get(a, 0, a.length)
```

Parameters:

`dst - The destination array`

Returns:

`This buffer`

Throws:

`BufferUnderflowException - If there are fewer than length floats remaining in this buffer`

put

```
public FloatBuffer put(FloatBuffer src)
```

Relative bulk *put* method *(optional operation)*.

This method transfers the floats remaining in the given source buffer into this buffer. If there are more floats remaining in the source buffer than in this buffer, that is, if `src.remaining() > remaining()`, then no floats are transferred and a `BufferOverflowException` is thrown.

Otherwise, this method copies *n* = `src.remaining()` floats from the given buffer into this buffer, starting at each buffer's current position. The positions of both buffers are then incremented by *n*.

In other words, an invocation of this method of the form `dst.put(src)` has exactly the same effect as the loop

```
while (src.hasRemaining())
    dst.put(src.get());
```

except that it first checks that there is sufficient space in this buffer and it is potentially much more efficient.

Parameters:

`src - The source buffer from which floats are to be read; must not be this buffer`

Returns:

`This buffer`

Throws:

`BufferOverflowException - If there is insufficient space in this buffer for the remaining floats in the source buffer`

IllegalArgumentException - If the source buffer is this buffer

ReadOnlyBufferException - If this buffer is read-only

put

```
public FloatBuffer put(float[] src,
                       int offset,
                       int length)
```

Relative bulk *put* method *(optional operation)*.

This method transfers floats into this buffer from the given source array. If there are more floats to be copied from the array than remain in this buffer, that is, if `length > remaining()`, then no floats are transferred and a BufferOverflowException is thrown.

Otherwise, this method copies `length` floats from the given array into this buffer, starting at the given offset in the array and at the current position of this buffer. The position of this buffer is then incremented by `length`.

In other words, an invocation of this method of the form `dst.put(src, off, len)` has exactly the same effect as the loop

```
for (int i = off; i < off + len; i++)
    dst.put(a[i]);
```

except that it first checks that there is sufficient space in this buffer and it is potentially much more efficient.

Parameters:

src - The array from which floats are to be read

offset - The offset within the array of the first float to be read; must be non-negative and no larger than array.length

length - The number of floats to be read from the given array; must be non-negative and no larger than array.length - offset

Returns:

This buffer

Throws:

BufferOverflowException - If there is insufficient space in this buffer

IndexOutOfBoundsException - If the preconditions on the offset and length parameters do not hold

ReadOnlyBufferException - If this buffer is read-only

put

`public final FloatBuffer put(float[] src)`

Relative bulk *put* method *(optional operation)*.

This method transfers the entire content of the given source float array into this buffer. An invocation of this method of the form `dst.put(a)` behaves in exactly the same way as the invocation

```
dst.put(a, 0, a.length)
```

Parameters:

`src - The source array`

Returns:

`This buffer`

Throws:

`BufferOverflowException - If there is insufficient space in this buffer`

`ReadOnlyBufferException - If this buffer is read-only`

hasArray

`public final boolean hasArray()`

Tells whether or not this buffer is backed by an accessible float array.

If this method returns `true` then the `array` and `arrayOffset` methods may safely be invoked.

Specified by:

`hasArray in class Buffer`

Returns:

`true if, and only if, this buffer is backed by an array and is not read-only`

array

`public final float[] array()`

Returns the float array that backs this buffer *(optional operation)*.

Modifications to this buffer's content will cause the returned array's content to be modified, and vice versa.

Invoke the hasArray method before invoking this method in order to ensure that this buffer has an accessible backing array.

Specified by:

array in class Buffer

Returns:

The array that backs this buffer

Throws:

ReadOnlyBufferException - If this buffer is backed by an array but is read-only

UnsupportedOperationException - If this buffer is not backed by an accessible array

arrayOffset

```
public final int arrayOffset()
```

Returns the offset within this buffer's backing array of the first element of the buffer *(optional operation)*.

If this buffer is backed by an array then buffer position *p* corresponds to array index *p* + `arrayOffset()`.

Invoke the hasArray method before invoking this method in order to ensure that this buffer has an accessible backing array.

Specified by:

arrayOffset in class Buffer

Returns:

The offset within this buffer's array of the first element of the buffer

Throws:

ReadOnlyBufferException - If this buffer is backed by an array but is read-only

UnsupportedOperationException - If this buffer is not backed by an accessible array

compact

```
public abstract FloatBuffer compact()
```

Compacts this buffer *(optional operation)*.

The floats between the buffer's current position and its limit, if any, are copied to the beginning of the buffer. That is, the float at index p = `position()` is copied to index zero, the float at index p + 1 is copied to index one, and so forth until the float at index `limit()` - 1 is copied to index n = `limit()` - 1 - p. The buffer's position is then set to $n+1$ and its limit is set to its capacity. The mark, if defined, is discarded.

The buffer's position is set to the number of floats copied, rather than to zero, so that an invocation of this method can be followed immediately by an invocation of another relative *put* method.

Returns:

`This buffer`

Throws:

`ReadOnlyBufferException - If this buffer is read-only`

isDirect

```
public abstract boolean isDirect()
```

Tells whether or not this float buffer is direct.

Specified by:

`isDirect in class Buffer`

Returns:

`true if, and only if, this buffer is direct`

toString

```
public String toString()
```

Returns a string summarizing the state of this buffer.

Overrides:

`toString in class Object`

Returns:

`A summary string`

hashCode

```
public int hashCode()
```

Returns the current hash code of this buffer.

The hash code of a float buffer depends only upon its remaining elements; that is, upon the elements from `position()` up to, and including, the element at `limit() - 1`.

Because buffer hash codes are content-dependent, it is inadvisable to use buffers as keys in hash maps or similar data structures unless it is known that their contents will not change.

Overrides:

`hashCode` in class `Object`

Returns:

The current hash code of this buffer

See Also:

`Object.equals(java.lang.Object)`, `System.identityHashCode(java.lang.Object)`

equals

```
public boolean equals(Object ob)
```

Tells whether or not this buffer is equal to another object.

Two float buffers are equal if, and only if,

1. They have the same element type,
2. They have the same number of remaining elements, and
3. The two sequences of remaining elements, considered independently of their starting positions, are pointwise equal. This method considers two float elements `a` and `b` to be equal if `(a == b) || (Float.isNaN(a) && Float.isNaN(b))`. The values `-0.0` and `+0.0` are considered to be equal, unlike `Float.equals(Object)`.

A float buffer is not equal to any other type of object.

Overrides:

`equals` in class `Object`

Parameters:

ob - The object to which this buffer is to be compared

Returns:

true if, and only if, this buffer is equal to the given object

See Also:

`Object.hashCode()`, `HashMap`

compareTo

```
public int compareTo(FloatBuffer that)
```

Compares this buffer to another.

Two float buffers are compared by comparing their sequences of remaining elements lexicographically, without regard to the starting position of each sequence within its corresponding buffer. Pairs of `float` elements are compared as if by invoking `Float.compare(float,float)`, except that `-0.0` and `0.0` are considered to be equal. `Float.NaN` is considered by this method to be equal to itself and greater than all other `float` values (including `Float.POSITIVE_INFINITY`).

A float buffer is not comparable to any other type of object.

Specified by:

`compareTo` in interface `Comparable<FloatBuffer>`

Parameters:

`that - the object to be compared.`

Returns:

`A negative integer, zero, or a positive integer as this buffer is less than, equal to, or greater than the given buffer`

order

```
public abstract ByteOrder order()
```

Retrieves this buffer's byte order.

The byte order of a float buffer created by allocation or by wrapping an existing `float` array is the `native order` of the underlying hardware. The byte order of a float buffer created as a view of a byte buffer is that of the byte buffer at the moment that the view is created.

Returns:

```
This buffer's byte order
```


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